

MacRay-WU4 Wi-Fi Ultrasonic Level Sensor Datasheet

Product Overview

MacRay-WU4 is a rugged **Wi-Fi-based ultrasonic level sensor** designed for short-range, high-precision level monitoring (0–4 meters) of liquids and solids in industrial, commercial, and utility environments where reliable IP connectivity is available. Using advanced ultrasonic time-of-flight measurement and intelligent echo processing, it delivers accurate and repeatable level data even in challenging conditions such as turbulence, foam, and confined spaces.

Engineered with an **IP67-rated sensor head**, corrosion-resistant housing, and industrial-grade electronics, MacRay-WU4 operates reliably in moisture-rich, dusty, and chemically exposed environments. It is well suited for tanks, pits, manholes, sumps, reservoirs, wastewater chambers, and flood-prone installations located within Wi-Fi-covered premises.

The device transmits level data securely over standard **Wi-Fi networks (2.4 GHz / 5 GHz)** directly to **on-premise servers, dashboards, SCADA systems, or cloud platforms** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**, without the need for gateways or long-range wireless infrastructure.

MacRay-WU4 supports **periodic, event-based, and threshold-based reporting**, along with **local BLE provisioning and diagnostics** via the Macnman Maya mobile application. Available in **battery-operated and externally powered variants**, it enables flexible deployment for both fixed industrial installations and locations where wired connectivity is limited, delivering low-latency, Wi-Fi-based level monitoring for modern Industrial IoT applications.





Key Features

Short-Range Precision Sensing

Optimized ultrasonic measurement for 0–4 meters liquid and solid level monitoring.

Battery-Powered LoRaWAN®

Multi-year autonomous operation with ultra-low-power wireless communication.

Smart Uplink Strategies

Periodic, event-based & threshold alerts optimize power consumption and network traffic.

Wide Industrial Use

Designed for manholes, sewer drains, tanks, pits, reservoirs & flood monitoring.

Flexible Deployment Options

Supports pole, wall, and custom mount installation for easy field deployment.

Advanced Echo Processing

Intelligent filtering eliminates false readings from foam, turbulence, and obstructions.

IP67 Rugged Design

Durable and corrosion-resistant enclosure built for harsh outdoor and underground environments.

BLE Configuration via Maya App

Quick setup, OTA parameter adjustment & diagnostics using a smartphone.

Secure & Long-Range Communication

AES-128 encrypted LoRaWAN® connectivity with strong penetration in remote areas.

Compatible with All LoRaWAN® Networks

Works seamlessly with private/public network servers and cloud platforms.

Suited Applications

Industrial Tank Level Monitoring

Accurate level measurement for water, chemicals, oils, and process liquids in industrial tanks located within Wi-Fi-enabled facilities.

Sumps, Pits & Utility Chambers

Continuous monitoring of liquid levels in sumps, lift pits, pump rooms, and underground utility chambers to prevent overflow and equipment damage.

Wastewater & Effluent Treatment Plants

Reliable level monitoring for equalization tanks, aeration basins, sludge pits, and treatment chambers within plant premises.

Flood & Basement Water Monitoring

Early detection of rising water levels in basements, parking areas, and flood-prone zones inside buildings and campuses.

Commercial Buildings & Campuses

Level monitoring for water tanks, rainwater harvesting systems, and drainage pits in offices, hospitals, malls, and campuses.

Water Storage & Distribution Systems

Level tracking in overhead tanks, underground reservoirs, and buffer tanks for industrial and commercial water management.

Industrial Process Automation

Integrates with SCADA and local dashboards over Wi-Fi for real-time level data used in automated control and safety systems.

Smart Factories & On-Premise IoT

Enables low-latency level monitoring within factories and plants using direct Wi-Fi connectivity and IP-based protocols.

Technical Specifications

General Specifications

Part Order Number	MacRay-WU4
Product Brand	Macnman Technologies Pvt Ltd

Ultrasonic Level Measurement

Ranging Range	3 – 450 cm
Blind Zone Distance	3 cm
Response Time (Sensor)	100 ms
Resolution	0.1 cm
Center Frequency	40 kHz ± 1 kHz
Sensing Angle	60°
Protection Rating (Sensor)	IP67

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client)

Access Point Mode (Configuration)

Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
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Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacRay-WU4 Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacRay-WU4-PO	12–24 V (Power)				Industrial or commercial sites with stable power supply
MacRay-WU4-BO	Battery	12 Ah			Remote or portable deployments with low power

What's in the Box?

Make sure box contains following items before installations



MacRay



Mounting Clip*



User manual & Warranty Card



IP Extender*



Mounting Screws

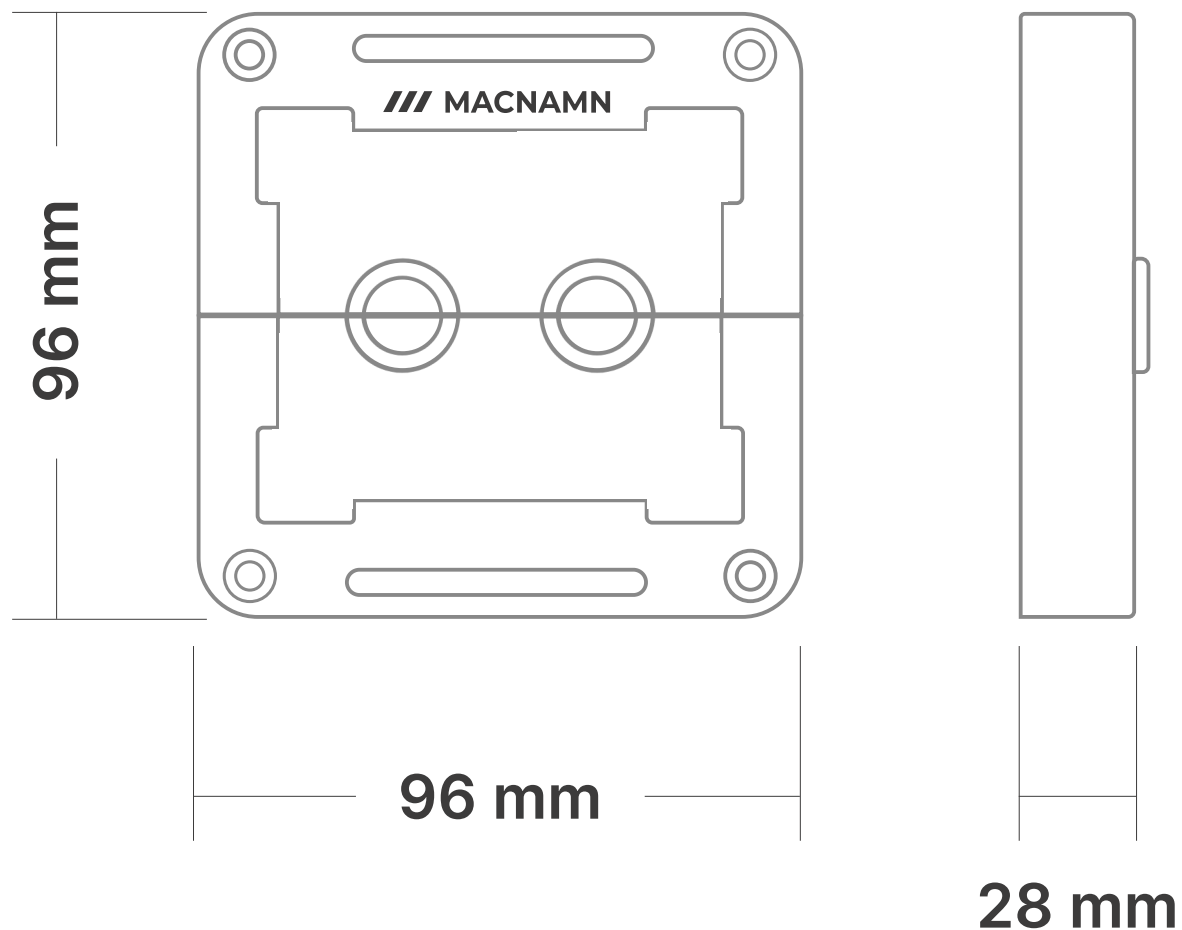


Power Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacRay-WU4 Dimensions



Frequently Asked Questions

What is the MacRay-WU4 used for?

MacRay-WU4 is a rugged Wi-Fi ultrasonic level sensor for short-range, high-precision (0-4 m) non-contact level monitoring of liquids and solids. Typical applications include Industrial Tank Level Monitoring, Sumps, Pits & Utility Chambers, Wastewater & Effluent Treatment Plants, Water Storage & Distribution Systems, Flood & Basement Water Monitoring and Industrial Process Automation.

What does the MacRay-WU4 measure, and how accurately?

Measurement specifications of the MacRay-WU4:

- Ultrasonic Level: range 3 – 450 cm, resolution 0.1 cm.

How does the MacRay-WU4 connect and send data over Wi-Fi?

The MacRay-WU4 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band). Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacRay-WU4 powered?

The MacRay-WU4 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacRay-WU4?

The MacRay-WU4 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacRay-WU4 configured?

The MacRay-WU4 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacRay-WU4 work with?

The MacRay-WU4 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacRay LUS X1 LoRaWAN ultrasonic level sensor](#), [MacRay-ToF-WXx Wi-Fi Time-of-Flight sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)